

Cloud Physics, Fall 2022

Homework 2

Homework Instructions

Due date: 2022/11/23 (Wed.) 10:00 AM

You should submit the following files to NTU COOL before the due date:

- A **pdf file** containing all your answers, including calculation and plots, if required by the question
- **Code(s)** used for calculation and/or plotting (if any)

Late Submission Policy:

- **20% deduction** between 11/23 10:00 AM ~ 11/30 10:00 AM
- **0 points** after 11/30 10:00 AM

Contact TA: b09209005@ntu.edu.tw

This homework contains three problems. Please answer them in the given order.

1. Supersaturation Development Equation (Chap.7)

$$\frac{ds}{dt} = A_1 \cdot \frac{dz}{dt} , \quad A_1 = 5.5 * 10^{-4} m^{-1}$$

Please calculate the **cloud base height (h)** under the following conditions, and show your calculation.

- (1) The saturation at $Z = 300$ m is 105.5 %, $h = ?$
- (2) The saturation at $Z = 500$ m is 102.0 %, $h = ?$
- (3) The saturation at $Z = 800$ m is 106.9 %, $h = ?$

[Hint]

- The saturation at cloud base is 100%.

2. Bowen's Model (Chap.8)

$$\frac{dR}{dt} = \frac{\bar{E} \cdot L}{4\rho_w} \cdot u(R)$$

$$\frac{dR}{dz} = \frac{dR}{dt} \frac{dt}{dz} = \frac{\bar{E} \cdot L}{4\rho_w} \cdot u(R) \cdot \frac{1}{w - u(R)}$$

$$\bar{E} = 1, L = 2.5 \text{ g m}^{-3}, u(R) = aR^b \text{ where } a = 8500 \text{ s}^{-1}, b = 1$$

Suppose the cloud base height is 200 m, and a droplet with initial radius r_0 starts in a cloud at height 300 m. The vertical speed of the cloud is w .

Consider the following four conditions:

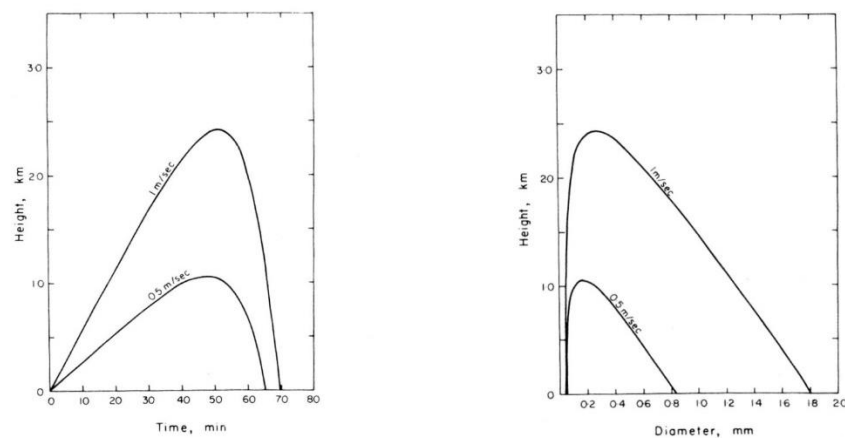
	w [m/s]	r_0 [μm]		w [m/s]	r_0 [μm]
A	2	15	C	4.5	15
B	2	40	D	4.5	40

With the given model and parameters, please:

- (1) Calculate the **radius** of the droplet(s) when it falls to the cloud base.
- (2) Plot the **Z-t diagram** and **Z-R diagram**
- (3) Compare the results and discuss.

[Hint]

- The plots should look similar to the following figure



- You may need to select a time step (dt) and apply numerical integration. This problem will require you to hand in your code.

3. Mason Equation (Chap.7)

$$R \frac{dR}{dt} = \frac{S - 1}{F_k + F_d}$$

$$\text{Suppose } \frac{1}{F_k + F_d} = 100 \mu m^2/s, \quad S = 75 \% = \text{constant}$$

Continue from problem 2, for the four conditions, calculate the **time** for the droplet(s) to fall from the cloud base to the surface ($Z = 0$), and the **radius** of the droplet(s) at the surface.

Compare the results and discuss.